Data Mining - UNIT-I

A **data warehouse** is a type of data management system that is designed to enable and support business intelligence (BI) activities, especially analytics. Data warehouses are solely intended to perform queries and analysis and often contain large amounts of historical data. The data within a data warehouse is usually derived from a wide range of sources such as application log files and transaction applications.

Data Warehouse

A data warehouse exhibits the following characteristics to support the management's decision-making process −

- **Subject Oriented** − Data warehouse is subject oriented because it provides us the information around a subject rather than the organization's ongoing operations. These subjects can be product, customers, suppliers, sales, revenue, etc. The data warehouse does not focus on the ongoing operations, rather it focuses on modelling and analysis of data for decision-making.
- **Integrated** − Data warehouse is constructed by integration of data from heterogeneous sources such as relational databases, flat files etc. This integration enhances the effective analysis of data.
- **Time Variant** − The data collected in a data warehouse is identified with a particular time period. The data in a data warehouse provides information from a historical point of view.
- **Non-volatile** − Nonvolatile means the previous data is not removed when new data is added to it. The data warehouse is kept separate from the operational database therefore frequent a change in operational database is not reflected in the data warehouse.

A data warehouse centralizes and consolidates large amounts of data from multiple sources. Its analytical capabilities allow organizations to derive valuable business insights from their data to improve decision-making. Over time, it builds a historical record that can be invaluable to data scientists and business analysts. Because of these capabilities, a data warehouse can be considered an organization's "single source of truth."

A typical data warehouse often includes the following elements:

- A relational database to store and manage data
- An extraction, loading, and transformation (ELT) solution for preparing the data for analysis
- Statistical analysis, reporting, and data mining capabilities
- Client analysis tools for visualizing and presenting data to business users
- Other, more sophisticated analytical applications that generate actionable information by applying data science and artificial intelligence (AI) algorithms, or graph and spatial features that enable more kinds of analysis of data at scale

Organizations can also select a solution combining transaction processing, real-time analytics across data warehouses and data lakes, and machine learning in one MySQL Database service—without the complexity, latency, cost, and risk of extract, transform, and load (ETL) duplication.

Benefits of a Data Warehouse

Data warehouses offer the overarching and unique benefit of allowing organizations to analyze large amounts of variant data and extract significant value from it, as well as to keep a historical record. Four unique characteristics (described by computer scientist William Inmon, who is considered the father of the data warehouse) allow data warehouses to deliver this overarching benefit. According to this definition, data warehouses are

- **Subject-oriented.** They can analyze data about a particular subject or functional area (such as sales).
- **Integrated.** Data warehouses create consistency among different data types from disparate sources.
- **Nonvolatile.** Once data is in a data warehouse, it's stable and doesn't change.

- **Time-variant.** Data warehouse analysis looks at change over time.

A well-designed data warehouse will perform queries very quickly, deliver high data throughput, and provide enough flexibility for end users to “slice and dice” or reduce the volume of data for closer examination to meet a variety of demands—whether at a high level or at a very fine, detailed level. The data warehouse serves as the functional foundation for middleware BI environments that provide end users with reports, dashboards, and other interfaces.

Difference between Operational database system and Datawarehouse

The Operational Database is the source of information for the data warehouse. It includes detailed information used to run the day to day operations of the business. The data frequently changes as updates are made and reflect the current value of the last transactions. Operational Database Management Systems also called as OLTP (Online Transactions Processing Databases), are used to manage dynamic data in real-time.

Data Warehouse Systems serve users or knowledge workers in the purpose of data analysis and decision-making. Such systems can organize and present information in specific formats to accommodate the diverse needs of various users. These systems are called as Online-Analytical Processing (OLAP) Systems.

Data Warehouse and the OLTP database are both relational databases. However, the goals of both these databases are different.

Operational Database	Data Warehouse
Operational systems are designed to support high-volume transaction processing.	Data warehousing systems are typically designed to support high-volume analytical processing (i.e., OLAP).
Operational systems are usually concerned with current data.	Data warehousing systems are usually concerned with historical data.
Data within operational systems are mainly updated regularly according to need.	Non-volatile, new data may be added regularly. Once Added rarely changed.
It is designed for real-time business dealing and processes.	It is designed for analysis of business measures by subject area, categories, and attributes.
It is optimized for a simple set of transactions, generally adding or retrieving a single row at a time per table.	It is optimized for extent loads and high, complex, unpredictable queries that access many rows per table.
It is optimized for validation of incoming information during transactions, uses validation data tables.	Loaded with consistent, valid information, requires no real-time validation.
It supports thousands of concurrent clients.	It supports a few concurrent clients relative to OLTP.
Operational systems are widely process-oriented.	Data warehousing systems are widely subject-oriented
Operational systems are usually optimized to perform fast inserts and updates of associatively small volumes of data.	Data warehousing systems are usually optimized to perform fast retrievals of relatively high volumes of data.
Data In	Data Out
Less Number of data accessed.	Large Number of data accessed.
Relational databases are created for on-line transactional Processing (OLTP)	Data Warehouse designed for on-line Analytical Processing (OLAP)

Feature	OLTP	OLAP
Characteristic	It is a system which is used to manage operational Data.	It is a system which is used to manage informational Data.
Users	Clerks, clients, and information technology professionals.	Knowledge workers, including managers, executives, and analysts.
System orientation	OLTP system is a customer-oriented, transaction, and query processing are done by clerks, clients, and information technology professionals.	OLAP system is market-oriented, knowledge workers including managers, do data analysts executive and analysts.
Data contents	OLTP system manages current data that too detailed and are used for decision making.	OLAP system manages a large amount of historical data, provides facilitates for summarization and aggregation, and stores and manages data at different levels of granularity. This information makes the data more comfortable to use in informed decision making.
Database Size	100 MB-GB	100 GB-TB
Database design	OLTP system usually uses an entity-relationship (ER) data model and application-oriented database design.	OLAP system typically uses either a star or snowflake model and subject-oriented database design.
View	OLTP system focuses primarily on the current data within an enterprise or department, without referring to historical information or data in different organizations.	OLAP system often spans multiple versions of a database schema, due to the evolutionary process of an organization. OLAP systems also deal with data that originates from various organizations, integrating information from many data stores.
Volume of data	Not very large	Because of their large volume, OLAP data are stored on multiple storage media.
Access patterns	The access patterns of an OLTP system subsist mainly of short, atomic transactions. Such a system requires concurrency control and recovery techniques.	Accesses to OLAP systems are mostly read-only methods because of these data warehouses stores historical data.
Access mode	Read/write	Mostly write
Insert and Updates	Short and fast inserts and updates proposed by end-users.	Periodic long-running batch jobs refresh the data.
Number of records accessed	Tens	Millions
Normalization	Fully Normalized	Partially Normalized
Processing Speed	Very Fast	It depends on the amount of files contained, batch data refresh, and complex query may take many hours, and query speed can be upgraded by creating indexes.

Why have a separate datawarehouse?

- A major reason is to help promote high performance of operational databases and datawarehouse.
- An operational database is designed and tuned for tasks such as indexing and hashing using primary keys, searching for particular records and optimizing queries.
- Datawarehouse queries are often complex. These queries involve computation of large groups of data at summarized levels, they may require the use of special data organization, access and implementation methods based on multidimensional values.
- So processing OLAP queries in operational databases would degrade the performance of operational tasks.
- An operational database supports the concurrent processing of multiple transactions, concurrency control and recovery mechanism like locking and logging are required to ensure the consistency and robustness of transactions.
- An OLAP query needs read-only access of data records for summarization and aggregation. If concurrency control and recovery mechanisms are applied for such OLAP operations it would harm the execution of concurrent transactions and reduce the throughput of an OLTP system
- A data warehouse requires consolidation of data from heterogeneous sources, which results in high quality, clean and integrated data.
- As both the systems provide different functionalities and require different kinds of data .It is necessary to maintain them separately.

What is Data mining?

The process of extracting information to identify patterns, trends, and useful data that would allow the business to take the data-driven decision from huge sets of data is called Data Mining.

In other words, we can say that Data Mining is the process of investigating hidden patterns of information to various perspectives for categorization into useful data, which is collected and assembled in particular areas such as data warehouses, efficient analysis, data mining algorithm, helping decision making and other data requirement to eventually cost-cutting and generating revenue.

Data mining is the act of automatically searching for large stores of information to find trends and patterns that go beyond simple analysis procedures. Data mining utilizes complex mathematical algorithms for data segments and evaluates the probability of future events. Data Mining is also called Knowledge Discovery of Data (KDD).

Data Mining is a process used by organizations to extract specific data from huge databases to solve business problems. It primarily turns raw data into useful information.

Data mining involves an integration of techniques from multiple disciplines such as database and datawarehouse technology, statistics, machine learning, high performance computing, pattern recognition, neural networks, data visualization, information retrieval, image and signal processing and spatial or temporal data analysis.

Data mining is the process of discovering interesting knowledge from large amounts of data stored in databases, data warehouses or other information repositories such as relational database systems, transaction processing systems and file systems.

What kind of patterns can be mined?

Based on the data functionalities, patterns can be further classified into two categories.

1) Descriptive patterns.

2) Predictive patterns.

Descriptive patterns

It deals with the general characteristics and converts them into relevant and helpful information.

Descriptive patterns can be divided into the following patterns:

1) Clusters
2) Associations
3) Frequent patterns
4) Correlations
5) Class/concept

- **Class/concept description**: Data entries are associated with labels or classes. For instance, in a library, the classes of items for borrowed items include books and research journals, and customers' concepts include registered members and not registered members. These types of descriptions are class or concept descriptions.

- **Frequent patterns**: These are data points that occur more often in the dataset. There are many kinds of recurring patterns, such as frequent items, frequent subsequence, and frequent sub-structure.

- **Associations**: It shows the relationships between data and pre-defined association rules. For instance, a shopkeeper makes an association rule that 70% of the time, when a football is sold, a kit is bought alongside. These two items can be combined together to make an association.
- **Correlations**: This is performed to find the statistical correlations between two data points to find if they have positive, negative, or no effect.

- **Clusters**: This is the formation of a group of similar data points. Each point in the collection is somewhat similar but very different from other members of different groups.

2) **Predictive patterns**: - They perform inference on the current data in order to make predictions.It predicts future values by analyzing the data patterns and their outcomes based on the previous data. It also helps us find missing values in the data.

Predictive patterns can be categorized into the following patterns.

1) Outlier analysis

2) Evolution analysis

3) Regression

4) Classification

- **Classification**: It helps predict the label of unknown data points with the help of known data points**.** For instance, if we have a dataset of X-rays of cancer patients, then the possible labels would be **cancer patient** and **not cancer patient**. These classes can be obtained by data characterizations or by data discrimination.

- **Regression**: Unlike classification, regression is used to find the missing numeric values from the dataset. It is also used to predict future numeric values as well. For instance, we can find the behavior of the next year's sales based on the past twenty years' sales by finding the relation between the data.
- **Outlier analysis**: Not all data points in the dataset need to follow the same behavior. Data points that don't follow the usual behavior are called outliers. Analysis of these outliers is called outlier analysis. These outliers are not considered while working on the data.
- **Evolution analysis**: As the name suggests, those data points change their behavior and trends with time.
- In some cases users may have no idea regarding what kind of patterns in their data may be interesting, so they may like to search for several different kinds of patterns in parallel.so it is important to have a data mining system that can mine multiple kinds of patterns to accommodate different user expectations or applications.
- The different kinds of patterns that are extracted from various types of data are- Itemset, sequence, rule, graph and event.

Class/concept description :-

Class/Concept refers to the data to be associated with the classes or concepts.

For example, in a company, the classes of items for sales include computer and printers, and concepts of customers include big spenders and budget spenders. Such descriptions of a class or a concept are called class/concept descriptions. These descriptions can be derived by the following two ways −

- **Data Characterization** − This refers to summarizing data of class under study. This class under study is called as Target Class. The output of data characterization can be presented in various forms. Example:-Pie charts, bar charts, curves, multidimensional data cubes and multidimensional tables including crosstabs.
- **Data Discrimination** − It refers to the mapping or classification of a class with some predefined group or class. It is the comparison of the general features of target class data objects with the general features of objects from one or a set of contrasting classes. The target and contrasting classes can be specified by the user and the corresponding data objects retrieved through database queries.

Mining of Frequent Patterns

Frequent patterns are those patterns that occur frequently in transactional data. Here is the list of kind of frequent patterns −

- **Frequent Item Set** − It refers to a set of items that frequently appear together, for example, milk and bread.
- **Frequent Subsequence** − A sequence of patterns that occur frequently such as purchasing a camera is followed by memory card.
- **Frequent Sub Structure** − Substructure refers to different structural forms, such as graphs, trees, or lattices, which may be combined with item-sets or subsequences.

Association and correlation

- In data mining association and correlation are the key techniques for extracting patterns and relationships from large datasets. Association exposes the relationships between items. Association refers to **any relationship between two variables.**
- **Association** is a technique used in data mining to identify the relationships or co-occurrences between items in a dataset. It involves analyzing large datasets to discover patterns or associations between items, such as products purchased together in a supermarket or web pages frequently visited together on a website. Association analysis is based on the idea of finding the most frequent patterns or itemsets in a dataset, where an itemset is a collection of one or more items.

- Correlation measures the strength of the link between two variables.
- Correlation is **a statistical measure that expresses the extent to which two variables are linearly related** (meaning they change together at a constant rate). It's a common tool for describing simple relationships without making a statement about cause and effect.
- **Correlation Analysis** is a data mining technique used to identify the degree to which two or more variables are related or associated with each other. Correlation refers to the statistical relationship between two or more variables, where the variation in one variable is associated with the variation in another variable. In other words, it measures how changes in one variable are related to changes in another variable. Correlation can be positive, negative, or zero, depending on the direction and strength of the relationship between the variables.
- **For example,** we are studying the relationship between the hours of study and the grades obtained by students. If we find that as the number of hours of study increases, the grades obtained also increase, then there is a positive correlation between the two variables.
- On the other hand, if we find that as the number of hours of study increases, the grades obtained decrease, then there is a negative correlation between the two variables.
- If there is no relationship between the two variables, we would say that there is zero correlation.

Classification and regression for predictive analysis

- Classification:-A classification is **an ordered set of related categories used to group data according to its similarities**. It consists of codes and descriptors and allows survey responses to be put into meaningful categories in order to produce useful data. A classification is a useful tool for anyone developing statistical surveys.
- It is the process of finding a model (or function) that describes and distinguishes data classes or concepts, for the purpose of being able to use the model to predict the class of objects whose class label is unknown.
- Classification predicts categorical labels, prediction models, and continuous valued functions. It is used to predict missing or unavailable numerical data values rather than class labels.
- The classification process deals with problems where data can be divided into binary or multiple discrete labels.
- Regression is **a statistical technique that relates a dependent variable to one or more independent variables**. A regression model is able to show whether changes observed in the dependent variable are associated with changes in one or more of the independent variables.
- The process of finding a model or function for distinguishing the data into continuous real values instead of using classes or discrete values. It can also identify the distribution movement depending on the historical data.
- In regression we have to predict a continuous target variable using independent features.
- Here we have two types of problems linear regression and nonlinear regression.

What is Outlier in data mining

Whenever we talk about data analysis, the term outliers often come to our mind. As the name suggests, "outliers" refer to the data points that exist outside of what is to be expected. The major thing about the outliers is what you do with them. If you are going to analyze any task to analyze data sets, you will always have some assumptions based on how this data is generated. If you find some data points that are likely to contain some form of error, then these are definitely outliers, and depending on the context, you want to overcome those errors. The data mining process involves the analysis and prediction of data that the data holds. In 1969, Grubbs introduced the first definition of outliers.

Outliers are discarded at many places when data mining is applied. But it is still used in many applications like fraud detection, medical, etc. It is usually because the events that occur rarely can store much more significant information than the events that occur more regularly.

Any unusual response that occurs due to medical treatment can be analyzed through outlier analysis in data mining.

The data objects that do not comply with the general behavior or model of data are called outliers. An outlier is a data point that stands out a lot from other data points in a set.

Most data mining methods discard outliers as noise or exceptions.

The analysis of outlier data is called as outlier mining. Outliers may be detected using statistical tests or using distance measures in which objects that are substantial distance from any other cluster are considered outliers.

Major issues in Data Mining

Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various heterogeneous data sources. These factors also create some issues. Here in this tutorial, we will discuss the major issues regarding –

- Mining Methodology and User Interaction
- Performance Issues
- Diverse Data Types Issues

Mining Methodology and User Interaction Issues

It refers to the following kinds of issues −

- **Mining different kinds of knowledge in databases** − Different users may be interested in different kinds of knowledge. Therefore it is necessary for data mining to cover a broad range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction** − The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
- **Incorporation of background knowledge** − To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple levels of abstraction.
- **Data mining query languages and ad hoc data mining** − Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- **Presentation and visualization of data mining results** − Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.
- **Handling noisy or incomplete data** − The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.
- **Pattern evaluation** − The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

Performance Issues

There can be performance-related issues such as follows −

- **Efficiency and scalability of data mining algorithms** − In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms** − The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data

into partitions which is further processed in a parallel fashion. Then the result from the partitions is merged. The incremental algorithms, update databases without mining the data again from scratch.

Diverse Data Types Issues

- **Handling of relational and complex types of data** − The database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kind of data.
- **Mining information from heterogeneous databases and global information systems** − The data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

Data Mining and Society

Data Mining is the process of collecting data and then processing them to find useful patterns with the help of statistics and machine learning processes. By finding the relationship between the database, the peculiarities can be easily identified. Aggregations of useful datasets from a heap of data in the database help in the growth of many industries we depend in our daily life and enhance customer service. We can't deny the fact that we live in a world of data. From the local grocery store to detecting network frauds, data mining plays a significant role. Beyond the benefits, data mining has negative impacts on society like privacy breaches and security problems. This article shows both the positive and negative effects of data mining on society.

Positive effects of data mining on society

Data mining has influenced our lives whether we feel it or not. Its applications are widely used in many fields to reduce strain and time. It has also supplemented the life of humans. Let's see some of the examples.

- **Customer relationship management:** By using the techniques of data mining the company provides customized and preferred services for customers which provides a pleasant experience while using the services. By aggregating and grouping the data, the company can create advertisements only when needed and it can reach the right people who require the service. By targeting the customer, unwanted promotional activities can be avoided which saves a lot of money for the company. The customer also doesn't get annoyed when heaps of junk mails and messages are not sent. Data mining can also help in saving time and provide satisfaction to the customers.
- **Personalized search engines:** In the world of data and networks, our lives become intertwined with web browsers. They had obtained an inevitable place in our lifestyle, knowledge and so on. With the help of data mining algorithms, the suggestions and the order of websites are tailored according to the gathered information by summarizing it. Ranking the page according to the content, the no of visits also help the web browser to provide necessary results for the query given by the user. By giving a personalized environment, spam and misleading advertisements can be avoided. By data mining, frequent spam accounts can be identified and they are automatically moved into the spam folder. For e.g. Gmail has a spam folder where unwanted and frequent junk messages are placed instead of heaping the inbox. Web-wide tracking is a process in which the system keeps track of every website a user visits. By incorporating the DoubleClick mechanism in these websites, they can note the websites that have been visited. And personalized lifestyle, educational ads are made visible in that sites relevant to that user.
- **Mining in the health sector:** Data mining helps in maintaining the health and welfare of a human. The layers of data mining embedded in pharmaceutical industries help to analyze data, to establish relationships while creating and improving drugs. It also helps in analyzing the effects of drugs on patients, the side effects and outcomes. They also help in tracking the number of chronically diseased patients, ICU patients which help in reducing the overflow of admissions in hospitals. Some medicines can also cause side effects or other benefits regardless of what disease it treats. In such cases, data mining can largely influence the growth of the health sector.

- **E-shopping:** E- retail platforms are one of the fastest-growing major industries in the world. From books, movies, groceries, lifestyles everything is listed on online e-retail platforms. This cannot run successfully without the help of data mining and predictive analysis. By these techniques, cross-selling and holding onto regular customers have become possible. Data mining helps in announcing offers and discounts to keep the customers intact and to increase sales. By using the algorithms of data science, the e-commerce website can largely influence the customers using targeted ad campaigns which will surge the number of users as well as it provide satisfactory results to customers.
- **Crime prevention:** Data mining plays a huge role in the prevention of crimes and reducing fraud rates. In telecommunication industries, it helps in identifying subscription theft and super-imposed frauds. It also helps in the identification of fraudulent calls. By doing this, user security can be ensured and prevent the company from facing a huge loss. It also plays an important role in police departments for identifying key patterns in crime and predicting them. It also helps in identifying the unsolved crimes committed by the same criminal by establishing a relationship between previous and present datasets in the crime database. By extracting and aggregating data, the police department can identify future crimes and prevent them. It also helps in identifying the cause of crime and the criminal behind that. This application largely supports the safety of people.

Negative effects of data mining on society

- **The exploitation of data and discrimination:** By agreeing on the terms and conditions provided by a company, the company gets access to collect data of the customers. From age groups to economical status, the company profiles the customers. By customer profiling, they get to know the datasets of rich, poor, elder, or younger. Some unethical or devious companies offer low credits or inferior deals to the customer in an area where fewer sales rate is noted. For. e.g. An unethical company decreasing the credit scores in the loyalty card of people connected to a branch whose transactions are less. Sometimes while profiling customers, wrongly accusing a customer happens. Though he is faultless, his needs and comfort are denied. Even though the company declares the customer faultless after investigations, still the wrongly accused customer struggles mentally and this incident will negatively impact his life. Certain companies don't take the responsibility of securing the data of customers which makes the data vulnerable and causes privacy breaches.
- **Health-related ethical problem:** Using data mining techniques, the companies can extract data about the health problems of the employees. They can also relate the summarized dataset with the datasets from the past history of previous employees. By discovering the pattern of diseases and frequencies, the company chooses the specific insurance plans accordingly. But, there is a chance that the company uses this data while hiring new employees. Hence, they avoid hiring people with a higher frequency of sickness. Insurance companies collect this data so they can avoid policies with companies with a high risk of health issues.
- **Privacy breach:** Every single piece of data we enter into the database of the internet is indirectly under the control of data miners. When used for unethical purposes, the privacy of an individual is invaded. Certain companies use this data to filter the latent people but with the potential to become customers of that company. In this way, the company sends targeted advertisements and increase customer traffic. For e.g. In telecommunication industries, the call details of the customer are created to enhance business growth and to maintain low customer churn. But, the company uses the data selfishly for its growth and it leads to the exploitation of privacy. Thus every single piece of data given to the network stands for greater risk under the influence of data mining.
- **Manipulation of data and unethical problems:** There are circumstances when normal data provided by a customer or user becomes manipulative data. That is when a customer makes a promotion on social media, which means he has a good financial status in his growing business. Using such information, miners can obtain unethical data to gain profits or access. Spreading of false information through social media and erroneous opinions can mislead people because when data miners collect this information, they become facts and which leads to a scam. Also, by using predictive analytics and machine learning algorithms, the outcome of an event is predicted by the government which may fail sometimes and that will create a disastrous effect on the public. When

the prediction is based on unprompted unsafe sources, those predictions lead to severe losses and the company may fail.

- **Invasive marketing:** The junk advertisements that heap your mobile while using social media or other social platforms are the result of data mining. Targeted ads benefit both seller and customer and save time but when it gets intense and unethical, wrong products are forced through advertisements that may negatively influence the life of the user. From the browser histories to previously purchased items, the data is extracted and used to influence the user to buy other products sometimes which may be harmful. This aggressive technique will cause undesirable effects on the user. Every discovery or field has its own merits and demerits. A part of that application may help the human and a part may degrade the values and ethics of society. As a part of the society we live in, it is our duty to use the applications of technology following the rules and maintaining ethics. Industries, companies and marketing agents should respect the privacy of individual humans and should provide the space they need. When every single person out there takes responsibility for the proper handling of data, data mining would be a gift of technology that could build and ease our life in so many ways.

**********END OF UNIT I**********